

Creative Programming: Sound

Lecture 5 (24 Feb)

Narration

Table of contents

- Bounding Values
- The World of Sound
- Playing sounds
- Recording

Narration

Click once to enable audio

Mic level: 0.000

Narration

(Re)visiting slides?

You can jump to any slide with the menu to the (bottom) left.

Did you miss a slide or want to revisit?

Open the narration tab while studying to get an explanation of difficult slides.



Milestone 1:

- We will go through the requirements for the first milestone, which is due next week

Milestone Guidance

- If you need some feedback/help for the first milestone, contact me during one of the class assignments or breaks
- As always, you can also ask and discuss with the TAs on Thursdays

Narration

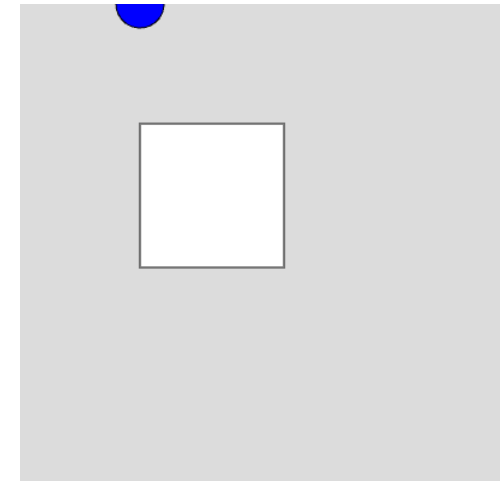
Bounding Values

Narration

Constraining via if

- if statements can be used to constrain values, but they can be verbose and error-prone.
- Example: constraining a variable `x` to be between 0 and 100.

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6   background(220);  
7   fill("white");  
8   rect(100, 100, 120, 120); // Box in the background  
9  
10  let constrainedX = mouseX; // We need to manipulate the value of mouseX, so we copy it to a  
    new variable  
11  
12 ▾ if (constrainedX < 100) { // Constrain left side  
13   constrainedX = 100;
```

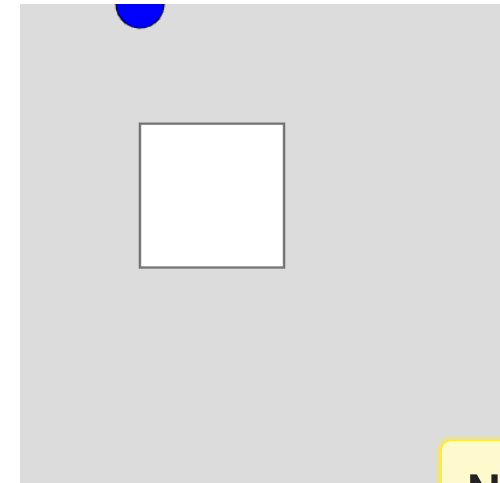


Narration

Constraining via constrain

- p5.js provides a built-in function `constrain()` that simplifies the process of constraining values.
- Example: constraining a variable `x` to be between 0 and 100 using `constrain()`.

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6   background(220);  
7   fill("white");  
8   rect(100, 100, 120, 120); // Box in the background  
9  
10  let constrainedX = constrain(mouseX, 100, 220); // Constrain mouseX between 100 and 220  
11  
12  fill("blue");
```

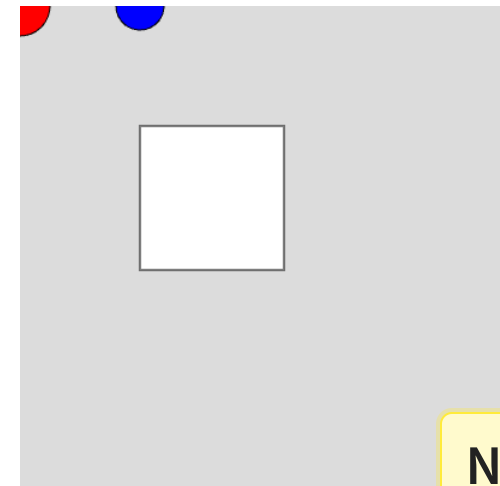


Narration

Constraining via min/max

- Another way to constrain values is by using the `min()` and `max()` functions.
- `min()` returns the smaller of two values, while `max()` returns the larger.
- Example: constraining a variable `x` to be between 0 and 100 using `min()` and `max()`.

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6   background(220);  
7   fill("white");  
8   rect(100, 100, 120, 120); // Box in the background  
9  
10  let constrainedX = min(mouseX, 220); // Constrain right side (100 + 120)  
11 }
```

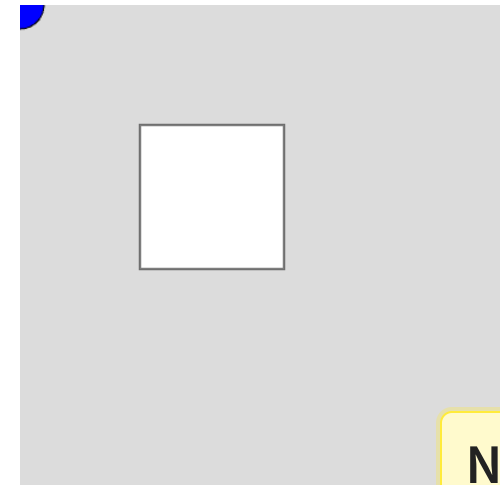


Narration

Constraining via modulo

- The modulo operator `%` can be used to wrap values around a certain range.
- This is particularly useful for creating looping behavior, such as in animations or games.
- Example: wrapping a variable `x` to be between 0 and 100 using modulo.

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6   background(220);  
7   fill("white");  
8   rect(100, 100, 120, 120); // Box in the background  
9  
10  let moduloX = (mouseX - 100) % 120 + 100; // Modulo to wrap mouseX between 100 and 220  
11 }
```

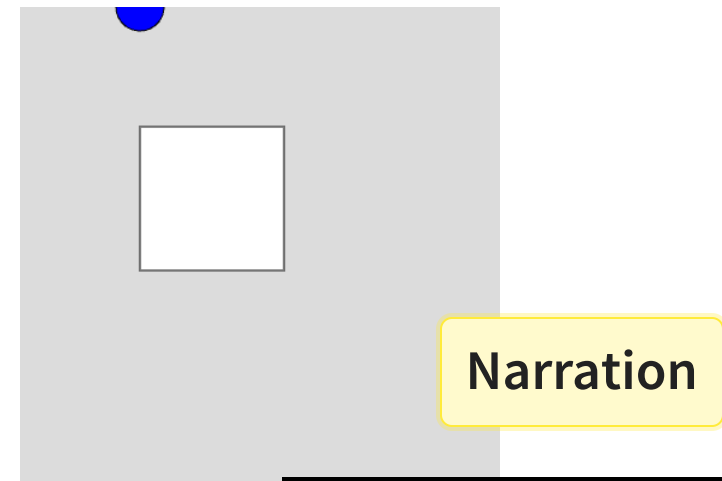


Narration

Constraining via mapping

- The `map()` function can be used to scale and constrain values from one range to another.
- You map two ranges to each other, and the function will automatically scale the input value to fit within the output range.
- Example: mapping a variable `x` from the range 0-255 to the range 0-100.

```
1 ▾ function setup() {  
2   createCanvas(400, 400);  
3 }  
4  
5 ▾ function draw() {  
6   background(220);  
7   fill("white");  
8   rect(100, 100, 120, 120); // Box in the background  
9 }
```



Constraining in action

💡 Class Assignment: Car

1. Create a shape that moves with the arrow keys. (Tip: check the reference!)
2. Use one of the constraining techniques to keep the shape within the canvas boundaries.
3. Can you make it so that the shape wraps around the canvas instead of stopping at the edges?



Narration

The World of Sound

Narration

Sound in the real world



Narration

Paint with Music

Paint with Music by Simon Doury
and Caroline Buttet



Narration

Coded Instruments



Narration

The synthesizer



Narration

The p5.js synthesizer

- We can play sound in p5.js using a synthesizer

```
1 let synthesizer;  
2  
3 function setup() {  
4   createCanvas(400, 400);  
5   synthesizer = new p5.MonoSynth()  
6 }  
7  
8 function draw() {  
9   background(220);  
10  text("Click me to play 'tuning A'", 100, height / 2);  
11  
12 }  
13  
14 function mousePressed() {  
15   userStartAudio();  
16   synthesizer.play("A4", 1, 0, 1);  
17 }
```

Narration

Libraries

- This code does not work by default! We need to include a library for extra functionality.
- The p5.js sound library provides various tools for working with sound.

Narration

Downloading the p5js sound library (1)

The screenshot shows the GitHub repository page for 'creative-programming-2026'. The repository is public and has 1 branch and 0 tags. The main branch is selected. The repository contains a folder named 'libs' with 376b808 commits, 3 minutes ago, and 45 commits. The file list includes:

File/Folder	Description	Time
lecture_1	Fix narration text	3 weeks ago
lecture_2	Fix narration text	3 weeks ago
lecture_3	fix palette code	2 weeks ago
lecture_4	lecture 4 updates	4 days ago
lecture_5/sound	yoga	1 hour ago
libraries	libs	3 minutes ago
README.md	reference link	4 days ago
reference.html	new lecture 4 link, reference	4 days ago

Narration

Go to the GitHub repository of the course and click on the **libraries** folder

Downloading the p5js sound library (2)

[creative-programming-2026](#) / [libraries](#) / 

 **Sipondo** libs

Name	Last commit message
 ..	
 p5.min.js	libs
 p5.sound.min.js	libs

Narration

Open the `p5.sound.min.js` file

Downloading the p5js sound library (3)

creative-programming-2026 / libraries / p5.sound.min.js



Sipondo libs 376b808 · 4 minutes ago History

Code Blame 3 lines (3 loc) · 195 KB

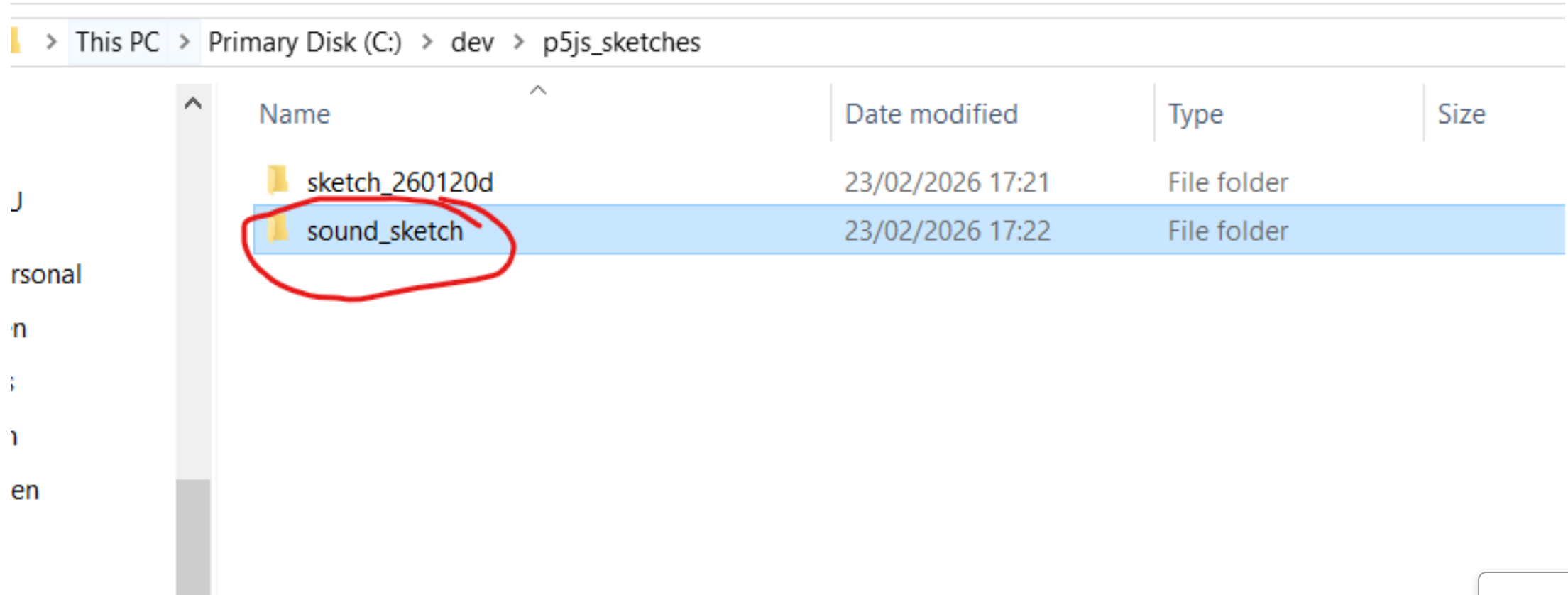
Download raw file

```
1  /** [p5.sound] Version: 1.0.1 - 2021-05-25 */
2  !function(n){var i={};function r(t){if(i[t])return i[t].exports;var e=i[t]={i:t,l:!1,exports:{}};return n[t].call(e.exports,e,e.exports,r),e.l=!0,e.exports}r.m=n,r.c=i,r.d=function(t,e,n){r.o(t,e)||Object.definePropertyProp
3  /**# sourceMappingURL=p5.sound.min.js.map
```

Download the raw file

Narration

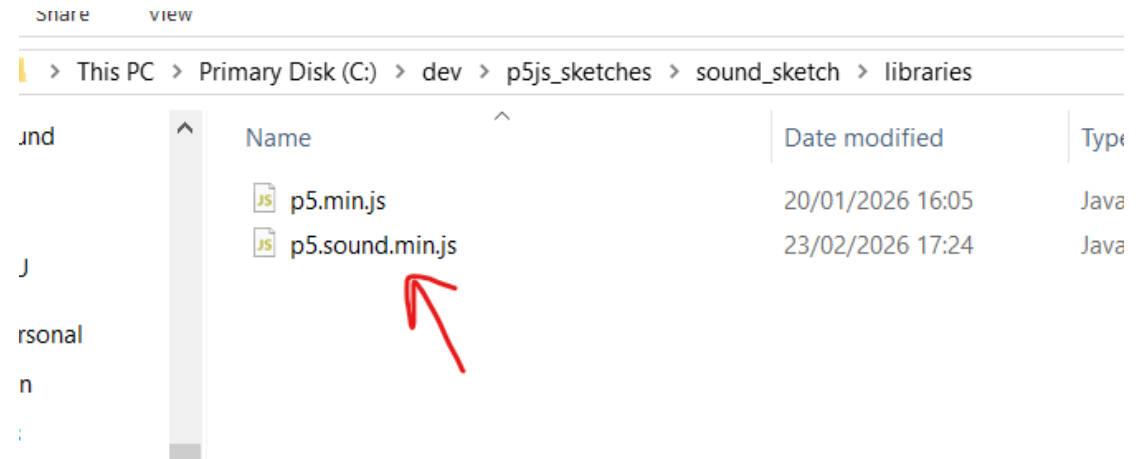
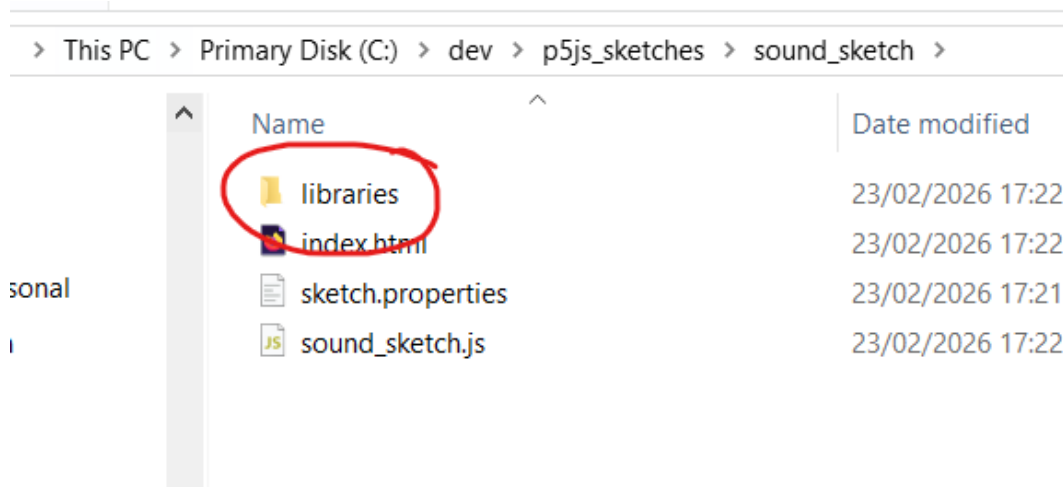
Putting the p5js sound library in your sketch (1)



Make your (new) sound sketch and go to its folder

Narration

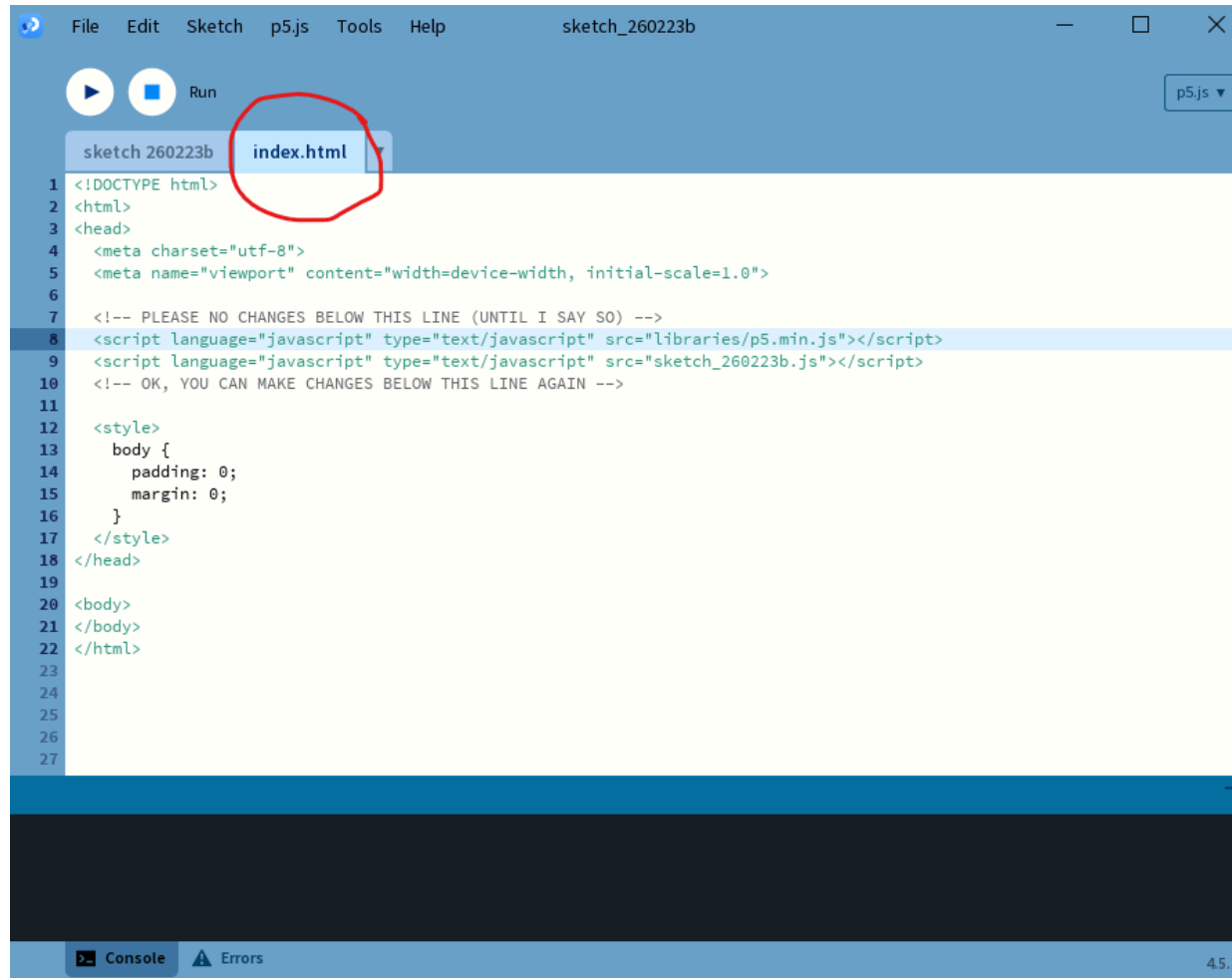
Putting the p5js sound library in your sketch (2)



Go to the libraries folder in your sound sketch and put the `p5.sound.min.js` file there

Narration

Setting up your sketch HTML (1)

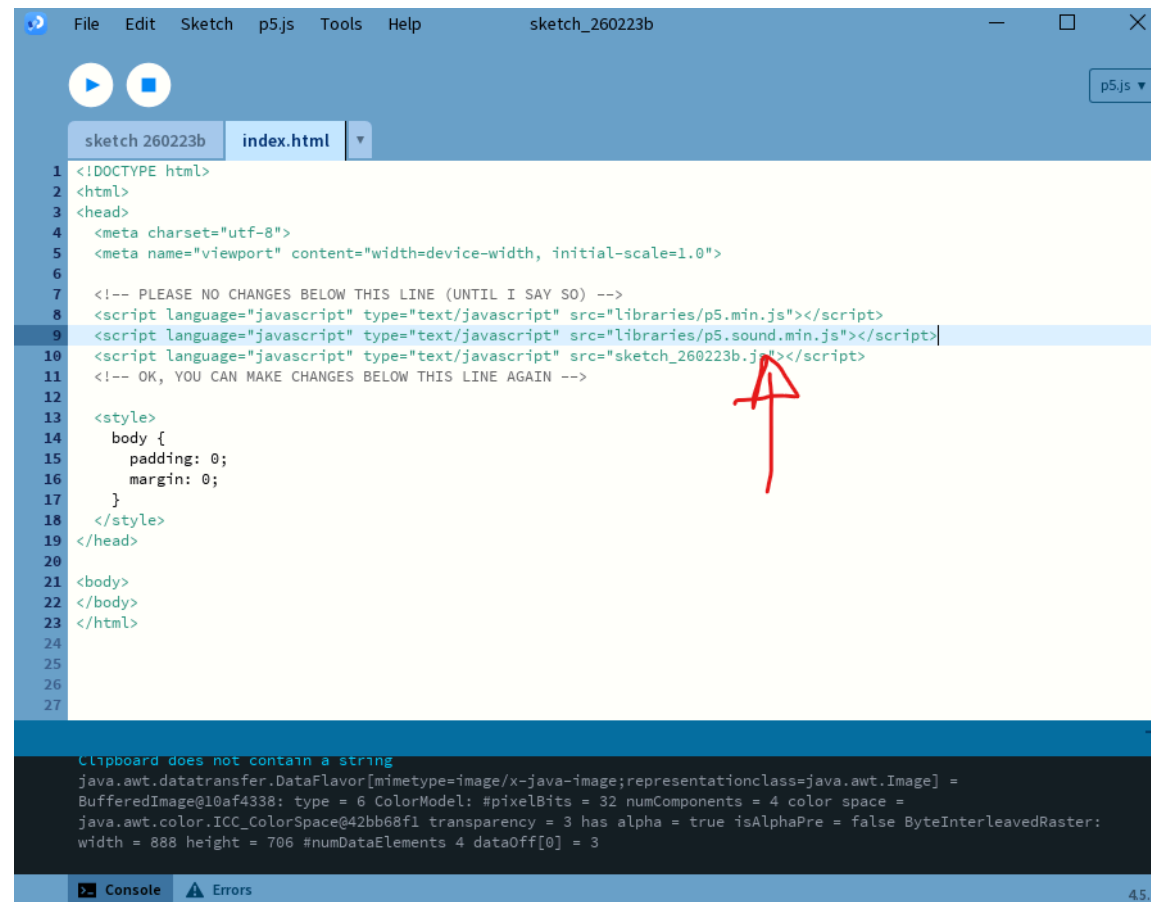


In Processing, open up the [index.html](#) file of your sketch

Setting up your sketch HTML (2)

Add the following to your index.html:

```
1 <script language="javascript" type="text/javascript" src="libraries/p5.sound.min.js"></script>
```



Narration

Setting up audio in your sketch

- Browser makers (Google, Mozilla, Apple, etc.) do not like it when websites play sound without the user's consent.
- To prevent this, they require that audio can only be played in response to a user interaction (e.g. a mouse click or key press).
- We have to call the `userStartAudio()` function in a user interaction event (e.g. `mousePressed()`) to enable audio in our sketch.

Narration

Playing a note

- We can play single notes by specifying their note name (e.g. “A4”) or their frequency in hertz (e.g. 440).

```
1  let mono;  
2  
3  ▾ function setup() {  
4    createCanvas(400, 400);  
5    mono = new p5.MonoSynth(); // Build the synthesizer. Don't forget "new"!  
6  }  
7  
8  ▾ function draw() {  
9    background(230);  
10   textSize(16);  
11   text("MonoSynth demo", 20, 40);  
12   text("Click to play", 20, 65);  
13   text("mouseX = pitch, mouseY = volume", 20, 85);  
14 }  
15  
16 ▾ function mousePressed() {
```

MonoSynth demo
Click to play
mouseX = pitch, mouseY = volume

Narration

Reacting to mouse position

- We can use frequencies to play notes that react to the mouse position.

```
1  let mono;  
2  
3  ▾ function setup() {  
4    createCanvas(400, 400);  
5    mono = new p5.MonoSynth(); // Build the synthesizer. Don't forget "new"!  
6  }  
7  
8  ▾ function draw() {  
9    background(230);  
10   textSize(16);  
11   text("MonoSynth demo", 20, 40);  
12   text("Click to play", 20, 65);  
13   text("mouseX = pitch, mouseY = volume", 20, 85);  
14 }  
15  
16 ▾ function mousePressed() {  
17   userStartAudio(); // Start the audio context on a user gesture (required in some browsers)  
18 }
```

MonoSynth demo
Click to play
mouseX = pitch, mouseY = volume

Narration

Playing multiple notes

- Multiple notes can be played with `p5.PolySynth()`
- Check out a website for this (e.g. chords) https://www.michael-thomas.com/music/class/chords_notesinchords.htm

```
1 let polySynth;
2
3 ▾ function setup() {
4   createCanvas(400, 400);
5   background(220);
6   textSize(20);
7   text('Click to play', 20, 20);
8
9   polySynth = new p5.PolySynth(); // Synth that can play multiple at once
10 }
11
12 ▾ function mousePressed() {
13   userStartAudio();
14 }
```

Click to play

Narration

Synthesizing!

💡 Class Assignment: Melody

1. Set up a synthesizer in p5js and play a note when a key is pressed.
2. Store a melody of notes in an array and play it back when a button is pressed instead.
3. By using a polySynth, can you play a melody of chords instead of notes?



Narration

Playing sounds

Narration

Sound files

- Of course we can also play sound files instead of synthesizing sounds ourselves.
- p5.js provides a `loadSound()` function to load sound files and a `play()` method to play them.
- You can stop, pause, and loop sounds as well using the `stop()`, `pause()`, and `loop()` methods.
- `setVolume()` sets the volume of the sound, similar to how we can set the volume of a synthesizer note.

Narration

The return of preload

- Just like with images, we need to use the `preload()` function to load sound files before the sketch starts.
- This ensures that the sound files are available when we want to play them.
- Remember! Loading files can be done with:
 - A website link (such as in the slides)
 - A local file in the same directory as your sketch (e.g. `cymbal.wav` if you have the file in the same folder as your sketch)

Narration

Sound example

- Let us look at an example of loading and playing sound files in p5.js.

```
1 let cymbal;
2 let drum;
3 let snare;
4
5 ▾ function preload() {
6   cymbal = loadSound( // Load the three sounds. You should prefer using local files, but here
   we load from a URL for ease of sharing
7     "https://rawcdn.githack.com/Sipondo/creative-programming-
2026/793c774ec9b9341473cd847312e7b6e21afcd410/lecture_5/sound/cymbal.wav"
8   );
9   drum = loadSound(
10    "https://rawcdn.githack.com/Sipondo/creative-programming-
2026/793c774ec9b9341473cd847312e7b6e21afcd410/lecture_5/sound/drum.wav"
11   );
12   snare = loadSound(
13    "https://rawcdn.githack.com/Sipondo/creative-programming-
2026/793c774ec9b9341473cd847312e7b6e21afcd410/lecture_5/sound/snare.wav"
14   );
```

Keys: C = cymbal, D = drum, S = snare

Narration

Finding sounds

- There are many websites where you can find free sound files to use in your projects, such as:
 - [Freesound](#)
 - [Free Music Archive](#)
 - [SoundBible](#)

Narration

Looping and pausing

- If you want to have a sound repeat, use `loop()` instead of `play()`.
- You can pause and resume the sound using `pause()` and `play()`.
- Finally, you can stop the sound completely with `stop()`.

```
1  let soundFile;
2
3  function preload() {
4    soundFile = loadSound("https://rawcdn.githack.com/Sipondo/creative-programming-
5    2026/7bcbf236e2ca4d161acb370092f669ce38cd543a/lecture_5/sound/yogabells.wav");
6  }
7  function setup() {
8    createCanvas(100, 100);
9    background(220);
10   text("Hold to play, release to pause", 10, 20, width - 20);
11 }
12
```

Hold to play,
release to
pause

Narration

Recording

Narration

Microphone input

- Sound goes both ways! We can also capture sound using the microphone.
- In p5js, we can use the `p5.AudioIn()` class to access the microphone and capture sound input.
- We have to make a new `p5.AudioIn()` object and call the `start()` method to begin capturing sound.

Narration

S(l)ide note

- Microphone input and interactive slides are not the best of friends.
- Some of the examples may not work properly in the slideshow, but you can copy the code into your own sketch to test it out.

Narration

Volume levels

- The `getLevel()` method of the audio input object returns the current volume level of the sound being captured by the microphone.
- This can be used to create visualizations or interactive elements that respond to sound input.

```
1  let mic;
2
3  function setup() {
4    createCanvas(400, 400);
5
6    // Create and start the microphone input
7    mic = new p5.AudioIn();
8    mic.start();
9  }
10
11 function mousePressed() {
12   userStartAudio(): // Start the audio context on a user aesture (required in some browsers)
```

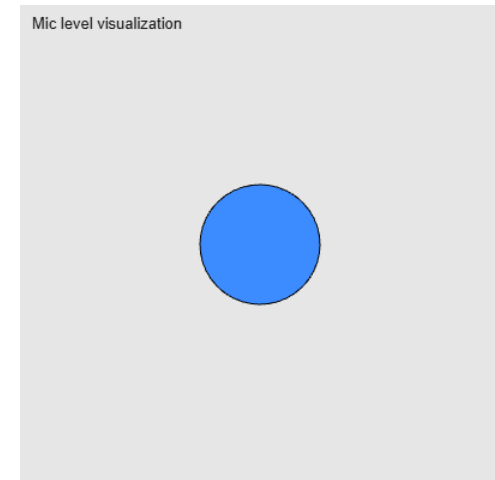
Mic level visualization

Narration

Mapping the microphone

- The values returned by `getLevel()` are typically between 0 and 1.
- We can use the `map()` function to scale the volume level from the microphone to a range that is suitable for our sketch.

```
1  let mic;
2
3  function setup() {
4    createCanvas(400, 400);
5
6    // Create and start the microphone input
7    mic = new p5.AudioIn();
8    mic.start();
9  }
10
11 function mousePressed() {
12   userStartAudio(); // Start the audio context on a user gesture (required in some browsers)
13 }
14
```

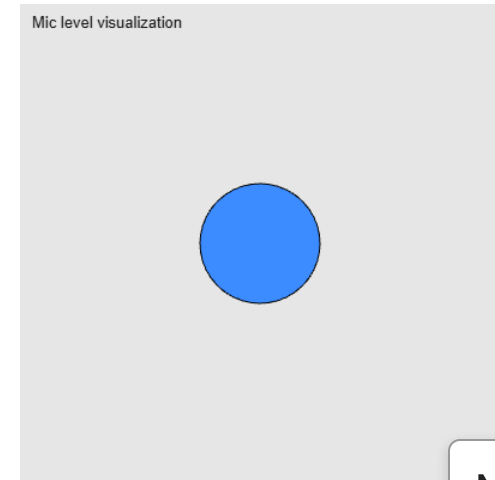


Narration

Easing the microphone

- The microphone values are very jittery and can change rapidly, which can make it difficult to create smooth visualizations or interactions.
- Easing can get rid of the jitter and create smoother transitions between values.

```
1  let mic;  
2  let diameter = 50; // Initial diameter for the circle  
3  
4  function setup() {  
5    createCanvas(400, 400);  
6  
7    // Create and start the microphone input  
8    mic = new p5.AudioIn();  
9    mic.start();  
10 }  
11  
12 function mousePressed() {
```



Narration

Sound as vibrations

- Microphone input does not just have to be sound
- Some media (such as games) use super high volume levels to check whether users blow into the microphone



Narration

Microphone-controlled synthesizer

- We can combine the microphone input with a synthesizer to create a sound-reactive instrument.
- For example, we can use the volume level from the microphone to control the frequency or amplitude of a synthesizer note.

```
1  let mono;  
2  let diameter = 50; // Initial diameter for the circle  
3  
4  ▾ function setup() {  
5    createCanvas(400, 400);  
6    mono = new p5.MonoSynth(); // Build the synthesizer. Don't forget "new"!  
7    mic = new p5.AudioIn(); // Create an audio input  
8    mic.start(); // Start the audio input  
9  }  
10  
11 ▾ function mousePressed() {  
12   userStartAudio(); // Start the audio context on a user aesture (required in some browsers)
```

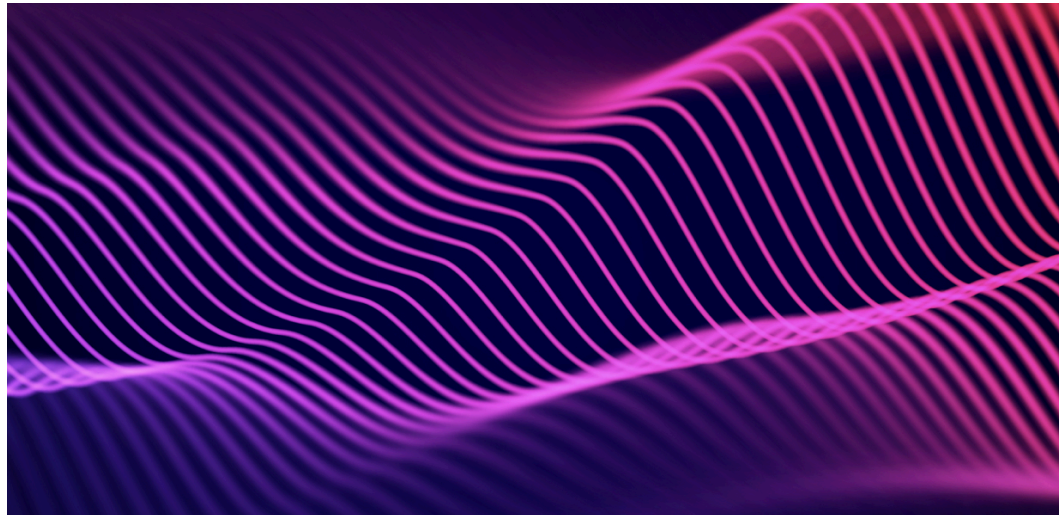
Click to play

Narration

Sound reactive

💡 Class Assignment: Sound Reactive

1. Set up a microphone input in p5.js and use the volume level to control a visual element (e.g. the size of a circle). You can also modify an old sketch!
2. Go on one of the free sound websites and find a sound file that you like. Load it into your sketch and play it when a button is pressed.
3. Can you make it so that the sound file loops as long as there is sound over the microphone?



Narration